

Spatial Analysis of Strain Distribution and Displacement in Japan Using ArcGIS and R: A Case Study of the March 11, 2011 Earthquake

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Abstract:

The March 11, 2011 Tohoku earthquake in Japan, a result of complex tectonic interactions, caused significant displacement across the country. This study aims to predict potential future seismic activity by analyzing strain distribution using data from over 900 Global Positioning System (GPS) stations. Data were retrieved from the University of Nevada, Reno, and processed using R. Displacement direction and magnitude were mapped using ArcGIS to explore strain distribution across the region. The analysis reveals a consistent displacement pattern in southern Japan, primarily directed southeast, while northern Japan experienced southern and western shifts. Closer to the epicenter, the GPS stations moved more unpredictably, seemingly compressing Japan together. These patterns align with the tectonic context, where the Okhotsk plate (OK) overrides the Pacific plate (PA), and the Philippine Sea plate (PS) subducts the Amur plate (AM). The results underscore the importance of continuous GPS monitoring and spatial distribution of strain in developing early warning systems for major seismic events.

Background Information:

- Japan is located at the convergence of three major tectonic plates: the Okhotsk Plate, the Amur Plate, and the Pacific Plate, making it highly seismically active.
- The interaction between the Okhotsk and Amur Plates is characterized by convergent and transform faults, which generate significant tectonic strain.
- GPS data shows displacement rates along the Okhotsk-Amur boundary typically range from 2 to 20 mm per year depending on any earthquake major event.
- Northern Japan: 4-7 mm/year on average.
- Near active fault zones: up to 10 or more mm/year.
- The accumulation of strain along these boundaries leads to seismic events when released.
- This research uses GPS displacement data to analyze strain patterns and identify potential earthquake frequency patterns.
- Advanced techniques such as spatial regression and Fourier analysis are employed to uncover these patterns, aiming to improve earthquake prediction capabilities.

Method/Materials:

- ArcGIS**
 - Creation of a GIS map of Japan
 - Projection of displacement rates arrows
- RStudio (v.4.2.1)**
 - Extrapolated 901 Gps time series data from geodesy.nwr.edu
 - Fill in missing data or replaced with interpolated using imputeTS
 - Moving average applied using kza package (Kolmogorov-Zurbenko)

$$Y_t = \frac{1}{m} \sum_{j=-k}^k (X_{t+j})$$

KZ-Filter equation utilized for the moving average adapted from Zurbenko (1986); package kza (Close et al. 2020).

- Downloaded time series GPS data using data.table package
- Fourier and spectral analysis using TSA package (periodogram function)

Results:

Data Collection: Processed thousands of geospatial data records from hundreds of GPS stations located in Japan to analyze tectonic activity.

Displacement Mapping: Developed displacement maps using R and ArcGIS to visualize strain along the boundary between the Okhotsk Plate and the Amur Plate.

During the 2011 major earthquake, the northeastern Japan region has experienced mostly south-west displacement with the highest displacement of ~1m, while the central Japan area shows more random displacement rates with a major components being directed both northwards and southwards.

Analytical Techniques: Employed spatial regression and Fourier analysis to study GPS displacement frequency patterns, particularly around the time of earthquakes.

Preliminary Findings: These frequency patterns indicate potential in predicting future earthquakes, though further research is needed to refine these predictions.

Impact: This research is an important step toward understanding earthquakes and potentially improving early warning systems in the future.

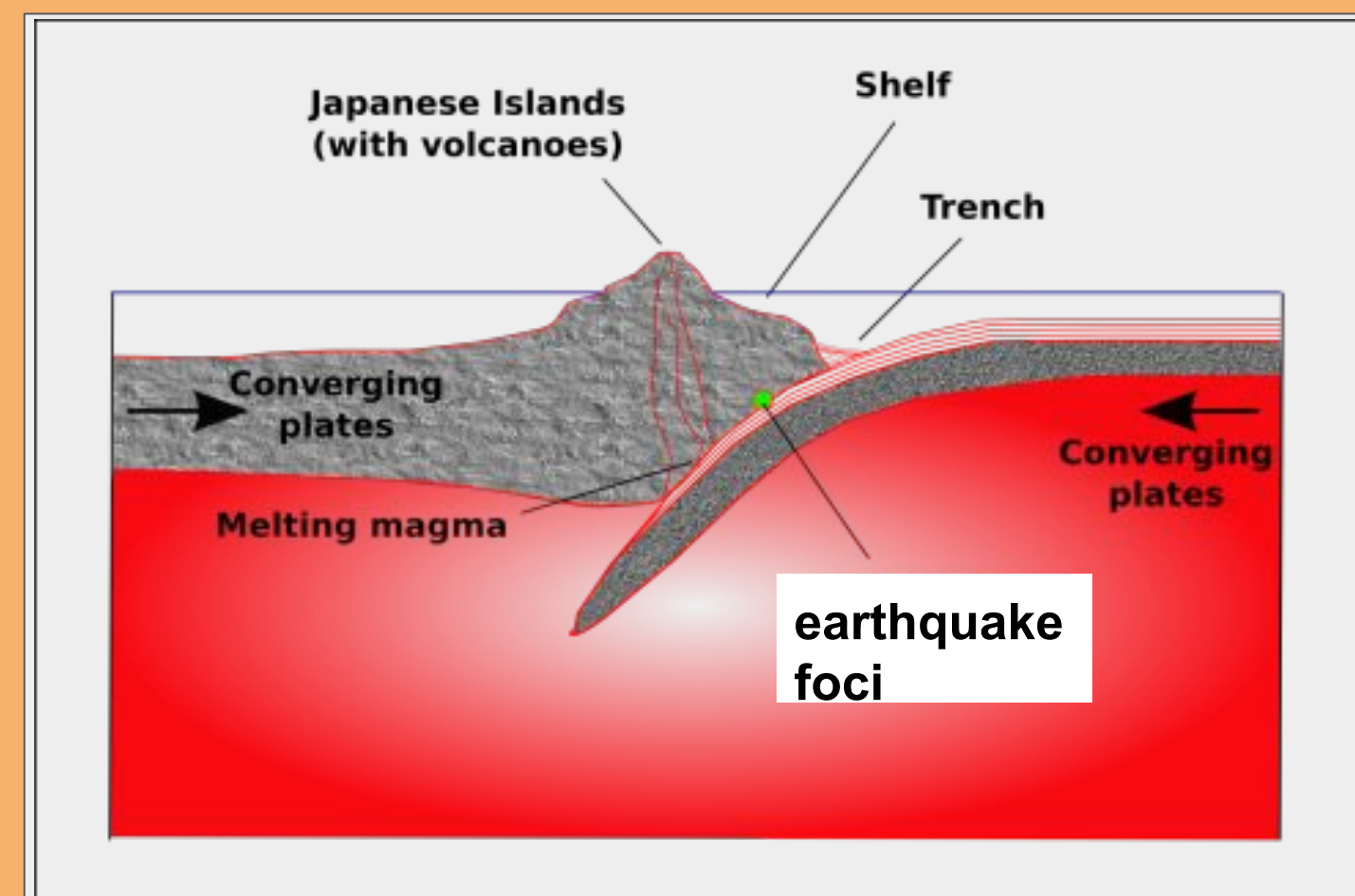


Figure 1: Conceptual figure showing the earthquake foci as the result of the interaction between the two tectonic plates in Japan.

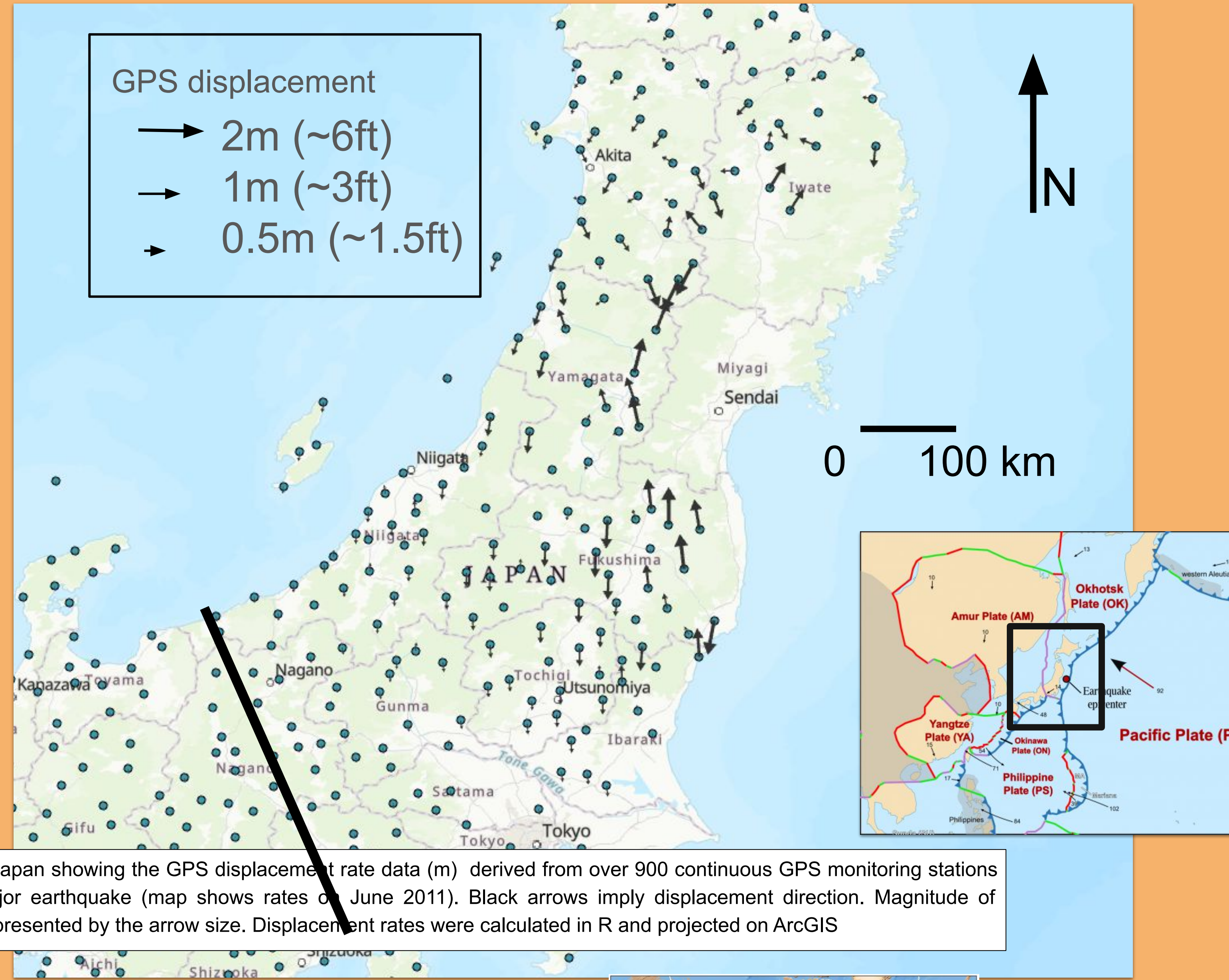


Figure 2: Map of Japan showing the GPS displacement rate data (m) derived from over 900 continuous GPS monitoring stations after the 2011 major earthquake (map shows rates of June 2011). Black arrows imply displacement direction. Magnitude of displacement is represented by the arrow size. Displacement rates were calculated in R and projected on ArcGIS

Figure 3: Global map of all GPS permanent stations showing the displacement rates with black arrows. The length of the GPS arrow denotes the intensity of annual displacement.

```
1 for(x in 1:999) {
2   gpsName <- ""
3   if(x < 10) gpsName <- paste0("J00", as.character(x))
4   else if(x < 100) gpsName <- paste0("J0", as.character(x))
5   else gpsName <- paste0("J", as.character(x))
6   tryCatch(
7     {
8       gpsData = GPS_dwd(gpsName)
9       startDate = "2024-06-01"
10      startIndex = which(gpsData$date == startDate)
11      endDate = "2024-07-01"
12      endIndex = which(gpsData$date == endDate)
13      diffNorth <- gpsData[endIndex, 2] - gpsData[startIndex, 2]
14      diffEast <- gpsData[endIndex, 3] - gpsData[startIndex, 3]
15      magVector <- sqrt(diffNorth^2 + diffEast^2)
16      bearing <- (90 - atan2(diffNorth, diffEast)*180/pi)%360
17      vectors <- rbind(vectors, list(gpsName, magVector, bearing))
18    },
19    error = function(cond) {
20      print("something went wrong")
21    }
22  )
23 }
```

Figure 4: Displacement rates and directions were calculated in Rstudio using a for loop with a customized function GPS_dwd() to download and process data from 901 continuous monitoring GPS stations.

Conclusion:

GPS displacement rates shown by the arrows further from the plate boundary between the two tectonic plates OK and AM reveals a ripple effect in land displacement rippling away from Sendai, the identified epicenter of the Mar 11 2011 Tohoku-oku earthquake.

Discussion:

- GPS displacement data along the Okhotsk-Amur Plate boundary has revealed strain patterns that may correlate with seismic events in Japan.
- Initial analysis using displacement maps, spatial regression, and Fourier analysis shows potential for identifying frequency patterns related to earthquakes.
- While promising, further research is required to refine these predictive models and explore their applicability to other tectonic boundaries.

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This research underscores the importance of geospatial monitoring and advanced analytical techniques for earthquake preparedness. We would like to thank Dr. Antonio E. Marsellos and the Geology, Environment and Sustainability department at Hofstra University and the HUSSEP for providing us with the necessary tools to conduct this research.

References:

- The long-term goal is to develop early warning systems that could provide crucial seconds of notice in earthquake-prone regions like Japan.
- Barrett T, Dowle M, Srinivasan A, Gorecki C, Chan K, Ripley B (2024). _kza_: Kolmogorov-Zurbenko Adaptive Filters. R package version 1.15.2. <https://CRAN.R-project.org/package=kza>.
- Bischi B, Lang M, Bossek J, Horn D, Richter J, Surmann D (2022). _BBmisc_: Miscellaneous Helper Functions for B. Bischi. R package version 1.13. <https://CRAN.R-project.org/package=BBmisc>.
- Chan K, Ripley B (2022). _TSA_: Time Series Analysis. R package version 1.3.1. <https://CRAN.R-project.org/package=TSA>.
- Close B, Zurbenko I, Sun M (2020). _kza_: Kolmogorov-Zurbenko Adaptive Filters. R package version 4.1.0.1. <https://CRAN.R-project.org/package=kza>.
- Garrett Grolemond, Hadley Wickham (2011). Dates and Times Made Easy with lubridate. Journal of Statistical Software, 40(3), 1-25. URL <https://www.jstatsoft.org/v40/i03/>.
- Blewitt, G., W. C. Hammond, and C. Kreemer (2018). Harnessing the GPS data explosion for interdisciplinary science, Eos, 99, <https://doi.org/10.1029/2018EO104623>. Also a map of the GPS continuous monitoring stations data: <http://geodesy.unr.edu/NGI/StationPages/gpsnetmap.GPSNetMap.html>
- Moritz S, Bartz-Beielstein T (2017). "imputeTS: Time Series Missing Value Imputation in R." _The R Journal_ *9*(1), 207-218. doi:10.32614/RJ-2017-009 <https://doi.org/10.32614/RJ-2017-009>.
- Ooms J (2024). _curl_: A Modern and Flexible Web Client for R. R package version 5.2.1. <https://CRAN.R-project.org/package=curl>.
- Tsakiri, K.G. & Marsellos, A.E., 2024. Signal detection in high-noise time series data using R. International conference on Time Series and Forecasting, ITISE2024, Spain, July 2024.
- Wickham H, François R, Henry L, Müller K, Vaughan D (2023). _dplyr_: A Grammar of Data Manipulation. R package version 1.1.4. <https://CRAN.R-project.org/package=dplyr>.